

Business Case

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Cloud-based ISMS

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Approval of the Business Case indicates an understanding of the purpose and content described in this deliverable. Approval of the Business Case constitutes approval of the business case analysis results and hereby certifies the overall accuracy, viability, and defensibility of the content and estimates. By signing this deliverable, each individual agrees the proposed business solution has been analyzed effectively as described herein. According to UAE Government Code, a state agency's executive director, or the executive director's designee, must approve.

Agency Head or Designee		
Nafiul Haque	nafnafiulhaque@gmail.com	0542415934
Signature: <i>Nafiul</i>		12/07/2024

Agencies may add additional signatories depending on internal project management governance.

Dr. Brinitha Raji	sraji@gbs.ac.ae	
Signature: <i>Brinitha</i>		14/07/2024

Momin Mohammed Nadeem	MYaseen@gbs.ac.ae	
Signature: <i>Nadeem</i>		13/07/2024

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Section 1. Project Justification

1.1 Business Problem

The United Arab Emirates-based real estate company Imtiyaz Development is encountering major challenges as a result of its extensive use of cloud technology. There are significant risks to data security, including breaches and unauthorized access, because the company handles sensitive data, including confidential manuscripts, design papers, invoices, payments, and customer information. Additionally, inconsistent data management across cloud services may result in operational inefficiencies and potential threats. The organization faces financial and legal risks due to non-compliance with international standards such as ISO 27001 and data protection laws in the United Arab Emirates.

The absence of a unified system for overseeing cloud security exacerbates these issues, leaving the company open to cyberattacks and complicating safe data access and cooperation. Putting in place a cloud-based Information Security Management System (ISMS) is essential to solving these issues. This solution will strengthen data security, guarantee adherence to legal requirements, and boost overall operational effectiveness.

1.2 Project Description

A hybrid development methodology that combines elements of waterfall and agile development will be used for the project. This hybrid approach will provide focused planning and execution, flexible adaptability to evolving needs and input, and iterative development.

Project Methodology:

1. Planning Phase

A. Requirement Analysis: Evaluate security requirements, regulatory compliance requirements, and Imtiyaz Development's present difficulties in-depth.

B. Project Plan Development: Draft a detailed project plan that includes the objectives, schedule, materials, and checkpoints.

2. Design Phase:

A. Solution Design: Create the cloud-based ISMS's architecture, taking into account data security measures, compliance features, and security controls.

B. Tool Selection: Select suitable SaaS products and tools that satisfy design specifications and work in unison with current cloud services.

3. Implementation Phase:

- A. System Configuration: Set up the chosen SaaS solutions in a way that complements the architecture that has been created.
- B. Integration: To guarantee seamless data flow and interoperability, integrate the ISMS with the current cloud services that Imtiyaz Development uses.

4. Testing Phase:

- A. Security Testing: To identify and fix vulnerabilities, carry out thorough security testing.
- B. Compliance testing: Verify that the system complies with all legal and regulatory standards.
- C. User Acceptance Testing (UAT): Test the system with important users and stakeholders to confirm its usability and functioning.

5. Deployment Phase:

- A. Rollout Strategy: To implement the ISMS without interfering with current activities, create a phased deployment strategy.
- B. Training: Provide employees with comprehensive training on how to use the new system and follow security procedures.

6. Maintenance and Moning Phase:

- A. Continuous Monitoring: To keep tabs on system security and performance, put continuous monitoring technologies into practice.
- B. Regular Updates: To keep the ISMS current with changing threats and compliance needs, schedule frequent updates and maintenance tasks.
- C. Feedback and Improvement: To create continuous system enhancements, and get input from users and stakeholders.

Project Methodology	Product Type
☐ Agile/Iterative ☐ Waterfall ☒ Hybrid ☐ Other ☐ Unknown/Unplanned	☐ Custom Development ☐ Legacy Migration ☒ Software as a Service (SaaS) ☐ Commercial off the Shelf ☐ Hybrid/Other (describe) ☐ Unknown/Unplanned

The proposed project aims to implement a cloud-based Information Security Management System (ISMS) to assist Imtiyaz Development in overcoming the major challenges that come with their substantial dependence on cloud technology. A uniform framework for managing data security and protection across several cloud services, strengthening data security, guaranteeing regulatory compliance, and boosting operational effectiveness will all be made possible by this ISMS.

1.2.1 QAT Best Practices

QAT Best Practices Considered	Agency will implement
☒ <i>Divide large projects up into less than \$10M smaller projects</i>	☒
☒ <i>Allocate adequate time to identify project requirements, procurement activities, and perform user-acceptance testing</i>	☒
☒ <i>DCS customers, engage DCS/STS team prior to posting solicitation</i>	☐
☒ <i>Leverage DIR's Shared Technology Services Program for project delivery needs related to cloud, application development, maintenance, security, and other technology solutions</i>	☒
☒ <i>Use of secure Open-Source software</i>	☒
☒ <i>Utilize agile development and user-centered design</i>	☒
☒ <i>Build IT systems using loosely coupled parts connected by open and available Application Programming Interfaces (APIs)</i>	☒
☒ <i>Include security planning throughout project lifecycle</i>	☒
☒ <i>Engage in IV&V services for projects over \$10M</i>	☐
☒ <i>Defer new scope to a later phase or follow-on project</i>	☐
☒ <i>Require remediation of system test defects and any performance-testing deficiencies before allowing project to proceed to the user-</i>	☒

<i>acceptance testing phase</i>	
☒ <i>Include network performance and capacity testing</i>	☒
☒ <i>Agile procurement</i>	☒
☒ <i>Include modular procurement</i>	☒
☒ <i>Assign a dedicated and empowered agency product owner</i>	☒

1.2.2 New or Modified Processes and Services

Processes/ Services	Description of Modifications/Automation	Users
Data Management	The project will implement a cloud-based Information Security Management System (ISMS) to enhance and automate data management processes. This system will safeguard sensitive data processing, storing, and sharing, including private manuscripts, design papers, invoices, payments, and customer information. The automation will include real-time data encryption, access control methods, and secure collaboration tools to reduce the risk of unauthorized access and data breaches.	Agency & Employers
Regulatory Compliance	The project will incorporate automated compliance monitoring tools into the ISMS to ensure that all data handling and security compliance procedures adhere to international standards like ISO 27001 and UAE data protection regulations. These technologies will lessen the financial and legal risks associated with non-compliance by regularly inspecting the system, generating compliance reports, and alerting the team to any deviations from essential standards.	Agency

Processes/ Services	Description of Modifications/Automation	Users
Authentication and User Access	Access control and user authentication will be automated via an improved user access and Agency Authentication authentication system integrated into the ISMS. This solution will combine role-based access control (RBAC), multi-factor authentication (MFA), and continuous monitoring to ensure that only authorized staff have access to sensitive data and systems, hence improving overall employer security.	Agency & Employers
Incident Response	Prompt security incident detection, analysis, and reaction are made possible by automated incident response capabilities, such as Agency reaction inside the ISMS. Automated workflows will be implemented in order to efficiently oversee the processes of incident reporting, investigation, and resolution on behalf of Employee. This will provide prompt reaction times, reducing the impact of security breaches and improving incident management overall.	Agency & Employers
Training and Awareness	The program will automate the process of providing security awareness and training to employees on Agency Awareness. An e-learning platform will be integrated into the ISMS to ensure that all staff members are informed on security protocols, industry best practices, and how to use the new ISMS effectively. This portal will include certification programs, examinations, and ongoing training modules.	Agency & Employers

1.2.3 Assumptions

1. Imtiyaz Development will be able to smoothly connect the cloud-based ISMS with its current cloud services.
2. All employees and interested parties will work together and take an active role in the new

system's adoption and training.

3. All necessary resources and money will be accessible for the duration of the project.
4. The project will have access to knowledgeable advisors and specialists in cybersecurity.
5. Throughout the project, compliance requirements and regulatory standards will not change.
6. The SaaS companies will deliver consistent service updates and strong support.
7. There will be adequate and dependable internet access to enable cloud-based ISMS functions.

1.2.4 Constraints

1. The project's scope and pace may be impacted by a limited budget and financial resources.
2. Time constraints since compliance standards demand prompt implementation.
3. Possible opposition to change from the employees used to the current procedures.
4. Reliance on external SaaS suppliers for upgrades and system capabilities.
5. Challenges with transferring data from the old ISMS to the new one.
6. Ensuring that company operations continue throughout the system changeover phase.
7. Adherence to applicable legislation and data protection laws in the United Arab Emirates.

1.2.5 Risks

1. Integration Problems: There may be problems connecting the ISMS to the current cloud services.
2. Employee Resistance: The staff's opposition to implementing new procedures and frameworks.
3. Data Breaches: During the transition phase, there is a risk of data breaches.
4. Compliance Changes: Modifications made to regulatory requirements during a project.
5. Resource Availability: There is a shortage of qualified cybersecurity experts.
6. System Downtime: Business activities may be impacted by system installation downtime.
7. Budget Overruns: The possibility of going over budget because of unanticipated expenses.
8. Vendor Reliability: Reliance on SaaS suppliers' dependability and effectiveness.
9. Training Effectiveness: There is a chance that certain training courses won't be sufficient to guarantee correct application and adherence.

10. Technical Failures: The ISMS may have technical issues that result in data loss or security flaws.

1.3 Current and Proposed Environment

1.3.1 Current Software

Software Items	Description	Hosting (On Prem, DCS, SaaS, other)
Cloud Storage Service	The current cloud storage service's primary goal is to securely manage and keep a variety of data types, including product designs, sales catalogs, and confidential client information. Strong security measures are required for this service, which is crucial to Imtiyaz Development's daily operations, to prevent data breaches and unauthorized access.	SaaS
Collaboration tools	Using a range of online meeting platforms and document-sharing technologies, employees work on confidential manuscripts and design papers. Even though these technologies improve cooperation and communication, they need to be used in a more secure environment to prevent data breaches and preserve confidentiality.	SaaS
Payment processing system and invoice	This system handles the business's financial activities, including the processing of invoices and payments. Ensuring the security of this system is crucial to prevent fraudulent activities and unauthorized access to financial data.	SaaS
Customer Relationship Management (CRM) system	Through the management of all client, employee, and customer data, the CRM system assists the company in maintaining relationships and streamlining processes. Because it handles sensitive personal data, this system must abide by data protection rules and have strong security measures in place.	SaaS

Software Items	Description	Hosting (On Prem, DCS, SaaS, other)
Communication tools and Email	These channels are used within the company for communications with both internal and external partners. Securing these technologies is necessary to prevent sensitive data sent over email and other communication channels from being compromised by cyberattacks.	SaaS

1.3.2 Current Hardware

Hardware Items	Description	Hosted (DCS, On Prem, IaaS, other)
Servers	Currently utilized for data and application hosting	On Premises
Network Switches	Control agency-wide network traffic	On Premises
Desktop Computers	Used regularly by employees of the agency	On Premises
Laptops	Arrangements for distant and mobile workers	On Premises
Printers	Applied to document printing	On Premises
Storage Arrays	Keep user and application data stored	On Premises
Firewalls	Protect the network of the organization from outside threats.	On Premises

1.3.3 Proposed Hardware / Software

Hardware/ Software Items	Description	Hosting (On Prem, DCS, SaaS, other)
Virtual Servers	Combined virtual servers for effectiveness and scalability	IaaS (Cloud-based)
Endpoint Devices	Contemporary desktop computers and laptops with improved security characteristics	On Premises
Network Appliances	Improved security and performance with upgraded switches and routers	On Premises
Firewall Solution	A sophisticated threat prevention feature-rich next-generation firewall	On Premises
Cloud Storage Solution (AWS)	Safe cloud-based data storage that is expandable and easily available	IaaS (Cloud-based)
Collaboration tools (Teams)	Cloud-based tools for teamwork and remote work	IaaS (Cloud-based)
Enterprise Software (Microsoft Office)	A software suite that is integrated and used to manage processes and operations	On Premises

1.3.4 DCS/STS Consideration

DCS	☐ Backup as a Service ☐ Compute and Storage ☐ Disaster Recovery as a Service ☐ Mainframe Services ☐ Microsoft O365 Subscription Services ☐ Network and Security ☐ Print & Mail
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	☒ Public Cloud Services ☐ Remote File ☐ Salesforce.com ☐ Texas Imagery Services ☐ Wide Area Application Service ☐ Application Development ☐ Application Maintenance ☐ Application Rate Card
Managed Security Services	☐ Digital Forensics ☒ Endpoint Management System Systems ☐ Intrusion Detection and Prevention Systems ☐ Malware Detention and Prevention ☒ Managed Firewall and Web Application Firewall (WAF) Services ☐ Penetration Testing ☐ Risk and Cloud Compliance Assessments ☐ Security Incident and Response Management Services ☐ Security Information and Event Management (SEIM)
Texas.gov	☐ Texas.gov Application Services ☐ Texas.gov Payment Services
Texas Open Data Portal	☐ Texas Open Data Portal

1.4 Data Classification and Retention

According to Texas Government Code 2054.161, on initiation of an information resources technology project, including an application development project and any information resources projects described in this subchapter, a state agency shall classify the data produced from or used in the project and determine appropriate data security and applicable retention requirements under Section 441.185 for each classification.

Data	Yes/No	Agency Comment
Has all data in scope been identified?	Yes	Data of various types include papers, invoices, and customer data
Has all data in scope been classified?	No	The classifying process is ongoing
Are there data security controls or records retention requirements already established?	Yes	Retention rules require improvement; basic restrictions are in place
Have all data owners and/or stewards, custodians been identified?	Yes	Determined departmental guardians for various categories of data
Has a data life cycle plan been created?	No	A plan to handle data from generation to disposal is now being developed
Have you developed a formal data governance program to manage the data in scope?	No	Basic rules are being defined and are in process
Have you conducted a data maturity assessment for the data in scope?	No	An evaluation of existing data management procedures is proposed

1.5 Security

Data	Definition	Yes/No
Will this project use, store, or transmit Personally Identifiable Information (PII) or Sensitive Personal Information (SPI)?	Personally identifiable information (PII), or Sensitive Personal Information (SPI), as used in Texas and US privacy law and information security, is information that can be used on its own or with other information to identify, contact, or locate a single person, or to identify an individual in context.	Yes

Is the data used in this project governed by any of the following standards?

Data Standard	Description of Data Standard	Yes/No
Minimum Acceptable Risk Security and Privacy Controls for Exchanges (MARS-E, CMS)	https://www.cms.gov/CCIIO/Resources/Regulations-and-Guidance/#MinimumAcceptableRiskStandards	Yes
Health Insurance Portability and Accountability Act (HIPAA)	http://www.hhs.gov/hipaa/for-professionals/security/	No
Family Educational Rights and Privacy Act (FERPA)	https://studentprivacy.ed.gov/Security	No
Payment Card Industry Data Security Standard (PCI DSS)	https://www.pcisecuritystandards.org/pci_security/standards_overview	Yes
Internal Revenue Service Publication 1075 (IRS)	https://www.irs.gov/pub/irs-pdf/p1075.pdf	Yes
Criminal Justice Information Services (CJIS, FBI)	https://www.fbi.gov/services/cjis/cjis-security-policy-resource-center	No
Social Security Administration (SSA)	https://www.ssa.gov/dataexchange/security.html	Yes
FedRAMP	https://www.fedramp.gov/	Yes
TX-RAMP	https://dir.texas.gov/texas-risk-and-authorization-management-program-tx-ramp	No
Other (Describe)		

What is the risk to the agency, state, or constituents if the system has a security event?

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Type	Level	Description. Use the highest that may apply. Low / Moderate / High should be defined by the agency.	Risk (Enter Low, Medium, High)
Legal	Low	No or Low legal risk.	Medium
	Medium	Moderate legal risk.	
	High	Severe legal risk.	
Financial	Low	No or Low financial risk.	Medium
	Medium	Moderate financial risk.	
	High	Severe financial risk.	
Business Processes	Low	No or low impact to business processes in the event of a security breach.	Medium
	Medium	Moderate chance of impact to business processes.	
	High	High chance of impact to business processes.	
Data	Low	No or low chance of non-regulated, non-PII data loss or leakage.	High
	Medium	Moderate chance of non-regulated, non-PII, or confidential data loss or leakage.	
	High	Regulated, PII, or confidential data may be lost or leaked.	
Resources	Low	Low number of resources would be required to return to normal business operations.	Medium
	Medium	Moderate number of resources would be required to return to normal business operations.	
	High	Significant number of resources would be required to return to normal business operations.	

1.6 Major Project Milestones

Milestones/Deliverables	Target Date mm/dd/yy
QAT Approval of Business Case	14/07/2024
Project Start Date	02/10/2024
Project Plan Submitted to QAT	14/07/2024
Acquisition Plan Submitted to QAT (Projects valued \$10M and above)	30/10/2024
Solicitation Posted (If necessary)	14/11/2024
Vendor Contract Executed (if necessary)	02/12/2024
Go Live	02/02/2025
Project Stabilization/Warranty e.g., 30 days without major issues/impact period (optional)	03/03/2025
Project Completion	14/03/2025
Post Implementation Review of Business Outcomes (due within 6 months of Completion)	14/09/2025

1.7 Business Objectives and Expected Benefits

Business Objectives (What does the project need to accomplish?)	Expected Benefits	Project Outcome Measures (How will you know you succeeded?)	Outcome Results Timing (When are results available?)

Business Objectives (What does the project need to accomplish?)	Expected Benefits	Project Outcome Measures (How will you know you succeeded?)	Outcome Results Timing (When are results available?)
Employ secure cloud-based solutions for data management, document collaboration, and online meetings	Enhanced productivity and cooperation when working remotely	Percentage growth in productivity indicators (such as document sharing and meeting attendance)	After 3 months of installation.
Enhance ISO standard compliance and data security	Upgraded regulatory compliance and safeguarding of data	Compliance audit results demonstrating adherence to ISO standards	After ISO certification is obtained
Simplify the handling of payments and invoicing processing	Reduced processing time and increased financial transaction accuracy	Invoice processing time and error rates have decreased	Post-implementation quarterly evaluations
Improve customer and client data management abilities.	Enhanced customer support and client relationship management.	An increase in client retention rates and satisfaction ratings.	Annually, starting 6 months after implementation

1.8 Cost (Budget Estimate)

Informational Cost	\$5,000 - \$25,000
Capital Cost	\$15,000 - \$55,000
Total Project Cost	\$20,000 - \$90,000

1.9 Alternatives Analysis

Alternative Options	Reasons For Not Selecting Alternative
Status Quo	The present inefficiencies in data administration, document collaboration, and online meetings would persist if this project were not carried out. This can result in ongoing operational inefficiencies and security threats.
Commercial Off the Shelf (COTS) Software	The chosen solution must be tailored to Imtiyaz Development's unique business requirements, which may not be adequately satisfied by off-the-shelf software. Integration and customization costs might be too high.
Cloud Computing (i.e., Hosted Solution)	Selected over preserving on-premises infrastructure because of its affordability, scalability, and flexibility. Data security and compliance issues might need rigorous vendor selection and continuous oversight.
“As a Service” model (e.g., software as service, system as service, infrastructure as a service etc.)	Reducing maintenance and operating expenses by using Software as a Service (SaaS) or other service models is possible, but relying too much on outside providers may cause issues with data security and service availability.
Other State or Agency Solutions	Due to variations in Imtiyaz Development-specific operating demands, security constraints, or integration challenges, it was taken into consideration but ultimately rejected.
[Add other option considered, if applicable.]	The distinct demands of Imtiyaz Development's real estate activities could not be entirely satisfied by current systems, requiring the development of new ones that are suited to their particular requirements. Outdated technology and integration issues might also be contributing concerns.

1.10 Justification

Justification Summary – Check All that Apply
☐ State legislative mandate or regulation
☐ Federal mandate or regulation
☐ Avoidance of funding loss or payment of penalties
☐ Financial benefit (ROI) to the state/agency
☐ Replacement of legacy technology solution
☒ Security
☒ Service improvement for Texans
☐ Other (describe)

Imtiyaz Development must implement secure cloud-based solutions to improve client and stakeholder service delivery, boost operational efficiency, and ensure data security. With the use of cloud-based technologies for online meetings, document collaboration, and data management, the company anticipates significant increases in staff productivity and external partner cooperation. The project addresses significant security vulnerabilities associated with outdated technology and time-consuming data processing techniques. Secure cloud solutions will fortify data security protocols to guarantee adherence to ISO standards and legal requirements for real estate data management practices. Additionally, the initiative is in line with Imtiyaz Development's corporate goals of improving customer data management, expediting the processing of invoices, and attaining operational excellence.

These goals should result in better customer happiness and service, which will eventually benefit Texans by offering more effective and timely services. The organization hopes to attain long-term financial benefits through lower infrastructure expenses and better resource allocation by utilizing cloud computing and service models. In addition to replacing antiquated systems, this calculated investment in contemporary technology sets Imtiyaz Development up for future expansion and scalability in the cutthroat real estate sector. Overall, the suggested project is supported by its compliance with legal requirements, enhanced data security, better service delivery, and projected financial gains from cost reductions and operational efficiency.

Section 2. Revision History

Version	Date mm/dd/yy	Name	Description
1.0	12/07/2024	Cloud-based ISMS Solution	A business plan for Imtiyaz Development to implement a cloud- based ISMS solution

Section 3. Appendices